

Stage 1 Feature Extraction

Sliding windows (1024 pt)
10 Time + 10 Freq + 8 DE
→ 28-D vector



Stage 2 Teacher 1D-CNN

3×Conv [16,32,64]
4-head Self-Attention
48,708 params · 99.93%



Stage 3 VRM-KD Distillation

$\tau=4.0$, $\alpha=0.3$
KAN[28,16,4] Student
6,148 params · 7.9x compr.



Stage 4 NeuroPLC Compiler

Frontend > IR > Optimize > SCL
6-op IR · 6 opt passes
DA Safety Factor 8.5x



Stage 5 TIA Portal V21

S7-1200: 45.2 KB (90.4%)
0 errors · 0 warnings
MCP Openness API validated